

README

https://github.com/Bananarya/DSCI560_lab5

1. Run *requirements.txt* to install all required packages.
2. Change the environment setting in the [automation.py](#) and *reddit_scrape.py* to your own sql settings.

```
def cfg():  
    return {"host":os.getenv("MYSQL_HOST","localhost"),"user":os.getenv("MYSQL_USER","pmaadmin"),"password":os.ge
```

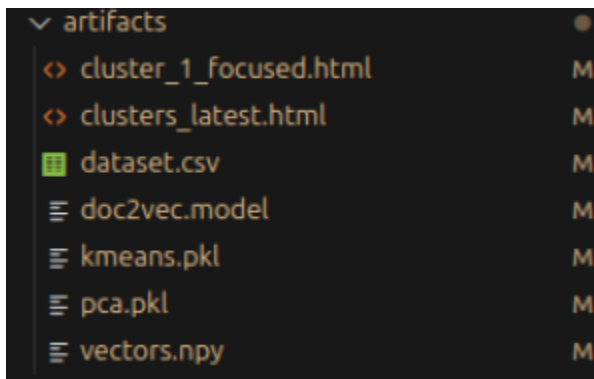
```
def main():  
    # Database configuration  
    db_config = {  
        'host': 'localhost',  
        'user': 'pmaadmin', # Replace with your MySQL username  
        'password': '20020718', # Replace with your MySQL password  
        'database': 'reddit_data'  
    }
```

3. Run [automation.py](#) by this syntax.

```
(myenv) bananarya@bananarya-VMware-Virtual-Platform:~/Desktop/DSCI560/DSCI560_la  
b5$ python3 automation.py  
usage: python automation.py <interval_minutes> [posts] [clusters]
```

For example, “*python3 [automation.py](#) 5 100 3*” means run the scraping and cluster program every 5 minutes, scraping 100 posts each time and classifying them into three clusters.

4. In the meanwhile, open another terminal, when the automation process completes the scraping and training process, you can run “*python3 [query.py](#)*” in another terminal and type in your message to see which cluster your message goes into and check all visualization in the artifact folder.



The screenshot shows a file explorer window with a folder named 'artifacts' expanded. Inside the folder, the following files are listed: 'cluster_1_focused.html', 'clusters_latest.html', 'dataset.csv', 'doc2vec.model', 'kmeans.pkl', 'pca.pkl', and 'vectors.npy'. Each file has a small icon to its left and a letter 'M' to its right.

The *cluster_1_focused.html* will show the visualization about your message and its cluster, the *cluster_latest.html* will show the visualization of the whole model's training data.